1.Aim/Overview of the practical: 1. Write an experiment to swap two columns in numpy.

2. Write an experiment to import a database with numbers and text keeping the text intact in python numpy.

2.Objective: The objective of this experiment is to load a structured data file (such as a CSV) that contains both numerical and textual data into a NumPy array while keeping the text data intact. NumPy is primarily designed for numerical data, so handling text data requires special consideration.

3.Procedure/Algorithm/Code:

1. import numpy as np

```
# Step 1: Create a sample NumPy array (3x3 matrix)
arr = np.array([[1, 2, 3],
                [7, 8, 9]])
print("Original Array:")
print(arr)
# Step 2: Swap column 0 (first column) with column 2 (third column)
arr[:, [0, 2]] = arr[:, [2, 0]]
print("\nArray After Swapping Column 0 and Column 2:")
print(arr)
```

2. import numpy as np

```
data = """ID,Name,Age,Salary
1,Nikhil,21,60000
2,varun,21,60000"""
with open('data.csv', 'w') as f:
    f.write(data)
data_array = np.genfromtxt('data.csv', delimiter=',', dtype=object, skip_header=1)


print("Loaded Data:")
print(data_array)
```

4.Output:

1.

Output

```
Original Array:
[[1 2 3]
 [7 8 9]]

Array After Swapping Column 0 and Column 2:
[[3 2 1]
 [9 8 7]]

[Execution complete with exit code 0]
```

2.

Output

```
Loaded Data:
[[b'1' b'Nikhil' b'21' b'60000']
 [b'2' b'varun' b'21' b'60000']]

[Execution complete with exit code 0]
```

Learning outcomes (What I have learnt):

1. **Understanding Mixed Data Types in NumPy.**
2. **File Handling and Data Parsing.**
3. **Array Indexing and Slicing in NumPy.**
4. **Efficient Data Manipulation.**